ALGEBRAIZING RIG-ETALE MORPHISMS OF AFFINE FORMAL SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-algebraize-rig-etale-affine; the proof environment immediately following it is its own proof

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Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the approximation lemma of the same chapter (lemma-approximate-by-etale-over-complement): a rig-etale morphism W to X_{/T} from a Noetherian affine formal algebraic space is realized as the completion of a finite type morphism etale over the complement
- the More on Morphisms lemma checking etale-ness on infinitesimal neighbourhoods (more-morphisms-lemma-check-smoothness-on-infinitesimal-nbhds)

Source reference. Original reference 0AKG: sources/stacks-restricted/pinned-original-source.tex, source lines 2406–2406. Prerequisite pointer invoked at 6418; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0A43: sources/stacks-chow/tags.txt, source lines 12761–12761. Prerequisite pointer invoked at 6421; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $T \subset X$ be a closed subset of a Noetherian affine scheme X. Let W be a Noetherian affine formal algebraic space. Let $g : W \to X_{/T}$ be a rig-étale morphism. Then there exists an affine scheme X' and a finite type morphism $f : X' \to X$ étale over $X \setminus T$ such that there is an isomorphism $X'_{/f^{-1}T} \cong W$ compatible with $f_{/T}$ and g. Moreover, if $W \to X_{/T}$ is étale, then $X' \to X$ is étale.*

Proof. The existence of X' is a restatement of Lemma 0AKG. The final statement follows from More on Morphisms, Lemma 0A43. □

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